

Standard attention:

$$R = \text{Softmax} \left(\frac{QK^T}{\sqrt{d}} \right) V$$

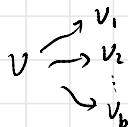
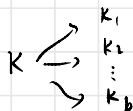
↪ Output

Suppose
that:

$$R = \begin{Bmatrix} R_1 \\ R_2 \\ \vdots \\ R_b \end{Bmatrix}$$

Flash attention:

We divide Q, K , and V into b small blocks, each block has t (queries, keys, or values)



Flash attention Compute output block $R_1, R_2, \dots, R_b$ Separately.

How does flash attention compute R_i gradually?
 We take R_i as an example, all other R_i can be updated similarly

$$R_1^{(1)} = \text{Softmax} \left(Q_1 (K_1^T) / \sqrt{d} \right) (V_1)$$

↪ update to

$$R_1^{(2)} = \text{Softmax} \left(Q_1 (K_1^T K_2^T) / \sqrt{d} \right) \begin{pmatrix} V_1 \\ V_2 \end{pmatrix}$$

↪ update to

$$R_1^{(3)} = \text{Softmax} \left(Q_1 (K_1^T K_2^T K_3^T) / \sqrt{d} \right) \begin{pmatrix} V_1 \\ V_2 \\ V_3 \end{pmatrix}$$

↪ update to

$\vdots$

↪ update to

$$R_1^{(b)} = \text{Softmax} \left(Q_1 (K_1^T \dots K_b^T) / \sqrt{d} \right) \begin{pmatrix} V_1 \\ \vdots \\ V_b \end{pmatrix}$$

after b steps,

$$R_1^{(b)} = R_i \text{ (which we want in the end)}$$

how do we update from $R_i^{(i)} \rightarrow R_i$ efficiently?

Let us take one query from Q_i as an example

Without loss of generality, we denote it as q_i .
we denote its output after step i as $r_i^{(i+1)}$

$$\begin{aligned}
 r_i^{(i+1)} &= \text{Softmax} \left(q_i^T \left(k_1, \dots, k_{(i+1)t} \right) / \sqrt{d} \right) \begin{pmatrix} v_1^T \\ \vdots \\ v_{(i+1)t}^T \end{pmatrix} \\
 &= \text{Softmax} \left((q_i^T k_1, \dots, q_i^T k_{(i+1)t}) / \sqrt{d} \right) \begin{pmatrix} v_1^T \\ \vdots \\ v_{(i+1)t}^T \end{pmatrix} \\
 &= \frac{\left(\exp \left(\frac{q_i^T k_1}{\sqrt{d}} - s^{(i)} \right) \dots \exp \left(\frac{q_i^T k_{(i+1)t}}{\sqrt{d}} - s^{(i)} \right) \right)}{\sum_{j=1}^{(i+1)t} \exp \left(\frac{q_i^T k_j}{\sqrt{d}} - s^{(i)} \right)} \begin{pmatrix} v_1^T \\ \vdots \\ v_{(i+1)t}^T \end{pmatrix} \\
 &= \frac{\sum_{j=1}^{(i+1)t} \exp \left(\frac{q_i^T k_j}{\sqrt{d}} - s^{(i)} \right) v_j^T}{\sum_{j=1}^{(i+1)t} \exp \left(\frac{q_i^T k_j}{\sqrt{d}} - s^{(i)} \right)} \\
 &= \frac{\sum_{j=1}^{it} \exp \left(\frac{q_i^T k_j}{\sqrt{d}} - s^{(i)} \right) v_j^T + \sum_{j=i+1}^{(i+1)t} \exp \left(\frac{q_i^T k_j}{\sqrt{d}} - s^{(i)} \right) v_j^T}{\sum_{j=1}^{(i+1)t} \exp \left(\frac{q_i^T k_j}{\sqrt{d}} - s^{(i)} \right)} \\
 &= \frac{\sum_{j=1}^{it} \exp \left(\frac{q_i^T k_j}{\sqrt{d}} - s^{(i)} \right) v_j^T}{\sum_{j=1}^{it} \exp \left(\frac{q_i^T k_j}{\sqrt{d}} - s^{(i)} \right)} \cdot \frac{\sum_{j=1}^{it} \exp \left(\frac{q_i^T k_j}{\sqrt{d}} - s^{(i)} \right) + \sum_{j=i+1}^{(i+1)t} \exp \left(\frac{q_i^T k_j}{\sqrt{d}} - s^{(i)} \right) v_j^T}{\sum_{j=1}^{(i+1)t} \exp \left(\frac{q_i^T k_j}{\sqrt{d}} - s^{(i)} \right)}
 \end{aligned}$$

$\swarrow$ $r_i^{(i)}$
 $\swarrow$ at each step, this sum can be updated
 $\swarrow$ at each step update this as well
 this depends only on block K_{i+1} and V_{i+1}